Introduction to Cache and Cache Block

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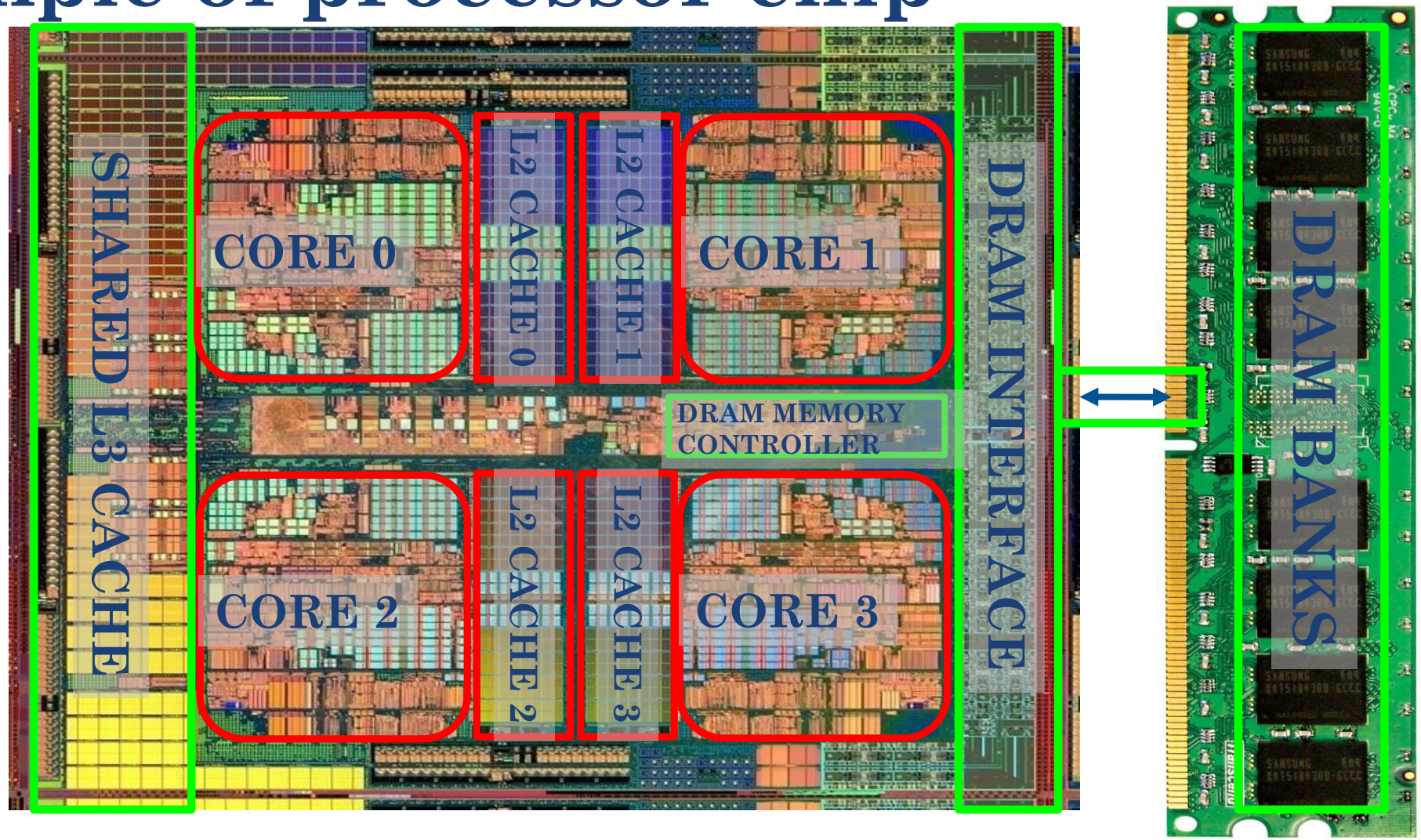
Courtesy for some slides: MK Publishers

Example of processor chip

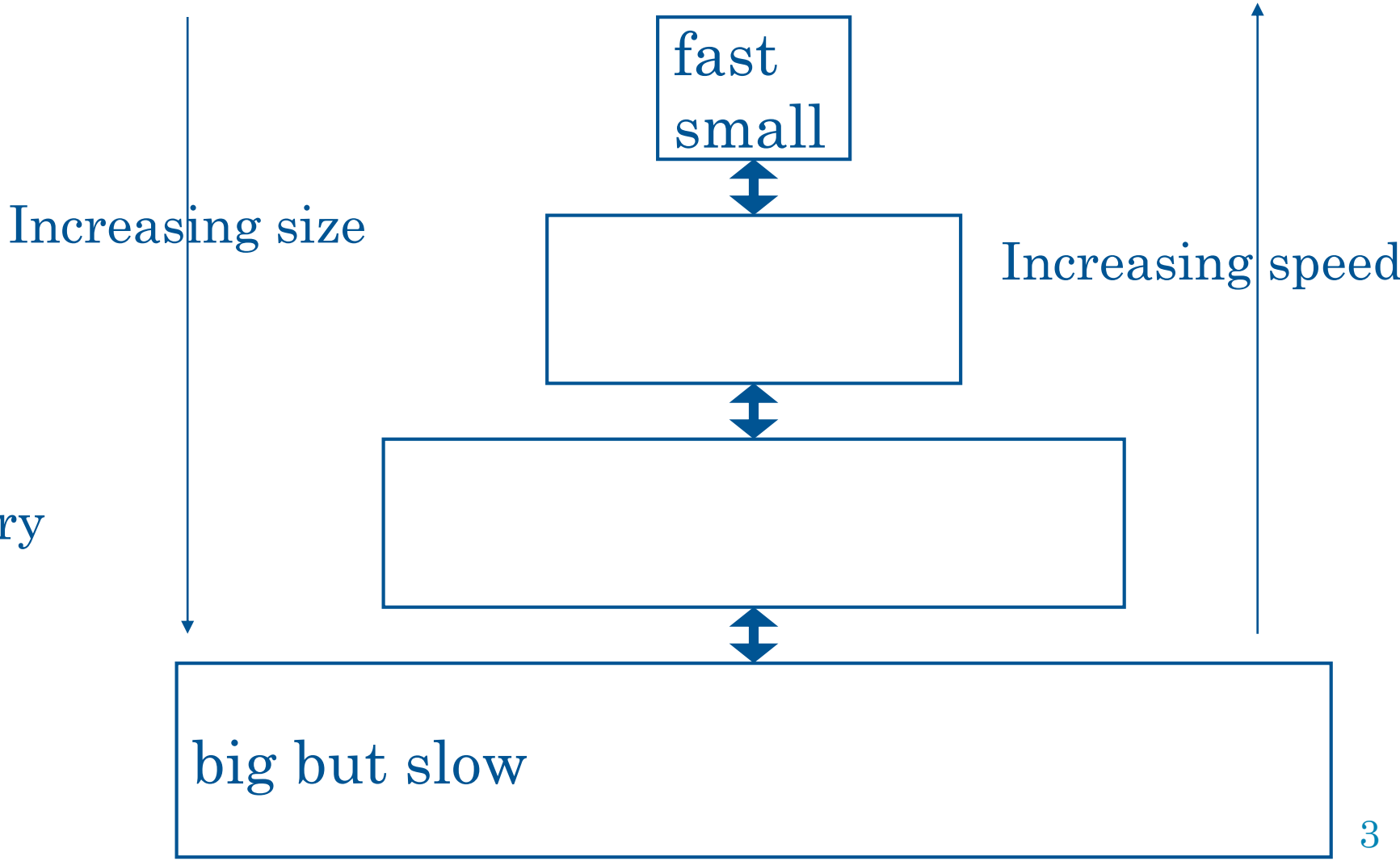


L1, L2, L3 cache => L1 is closest and L3 is farthest from core

Courtesy: Onur Mutlu

Why memory "hierarchy"?

Personal bag	L1 cache
Book-shelf	L2 cache
Institute library	Main memory
Inter-institute collection	Hard disk



Example of Memory Hierarchy

Register File
32 words, sub-nsec

L1 cache
~32 KB, ~nsec

L2 cache
512 KB ~ 1MB, many nsec

L3 cache,
.....

Main memory (DRAM),
GB, ~100 nsec

Disk
100 GB, ~10 msec

Courtesy: Onur Mutlu

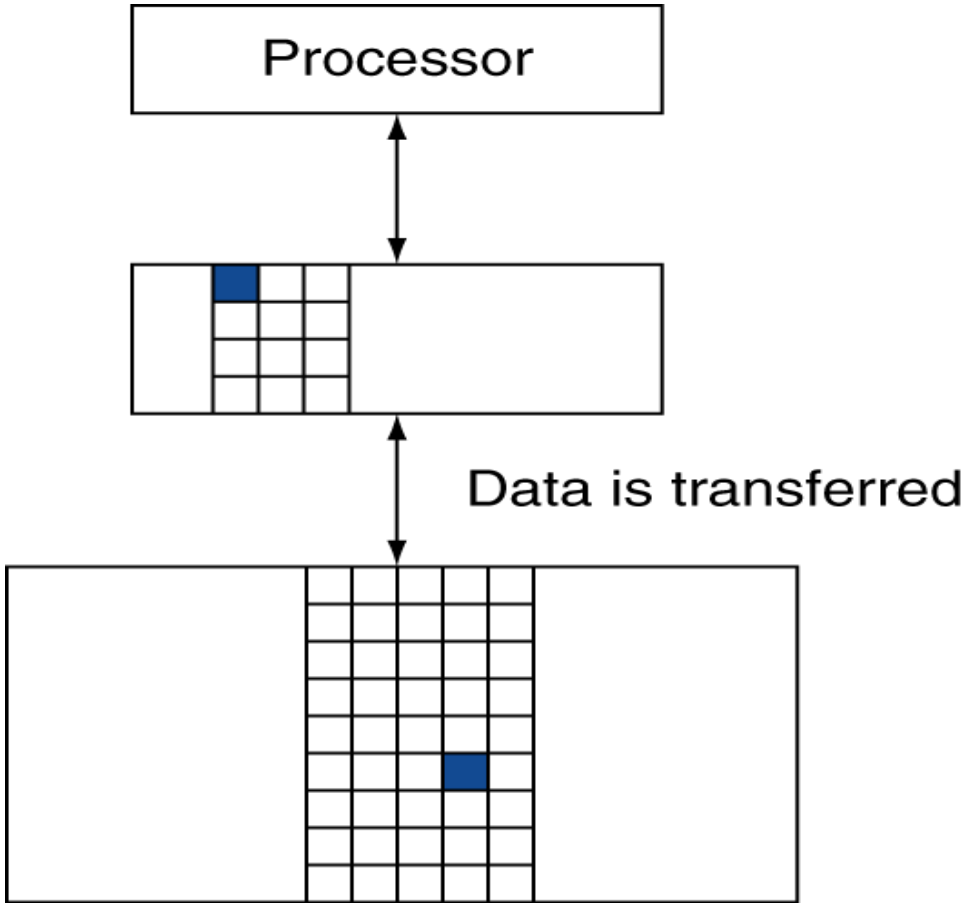
Principle of Locality

- Programs access a small proportion of their address space at any time
- **Temporal locality**
  - Items accessed recently are likely to be accessed again soon
  - e.g., instructions in a loop
- **Spatial locality**
  - Items near those accessed recently are likely to be accessed soon
  - E.g., sequential instruction access, array data

Memory Technology

- **Static RAM (SRAM)**
  - 0.5ns – 2.5ns
- **Dynamic RAM (DRAM)**
  - 50ns – 70ns
- **Magnetic disk**
  - 5ms – 20ms
- **Ideal memory**
  - Access time of SRAM
  - Capacity and cost/GB of disk

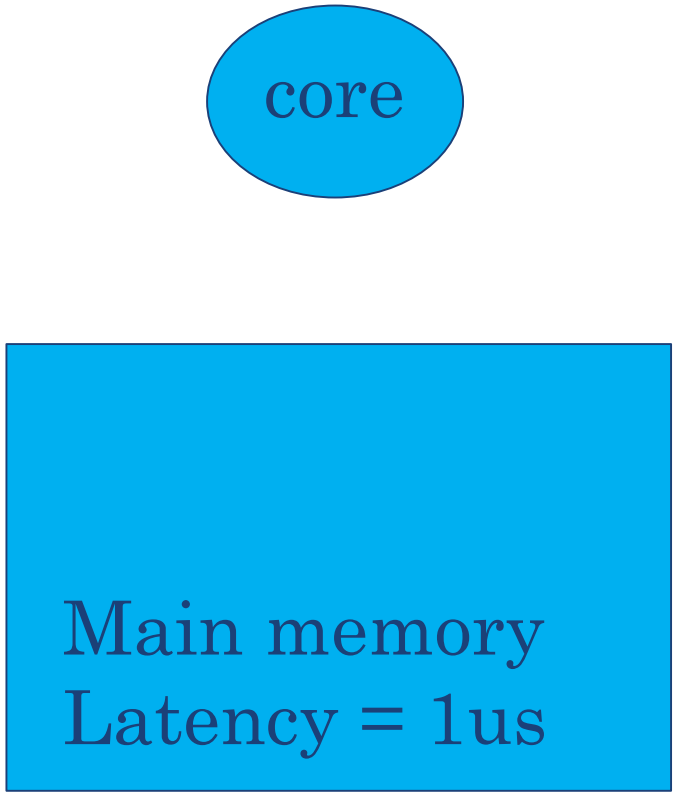
Memory Hierarchy Levels



- Block (aka line): unit of copying
- If accessed data is present in upper level
  - Hit: access satisfied by upper level
- If accessed data is absent
  - Miss: block copied from lower level
  - Then accessed data supplied from upper level

Patterson, David A., and John L. Hennessy. "Computer organization and design: the hardware/software interface, 2012.

Why memory "hierarchy"?

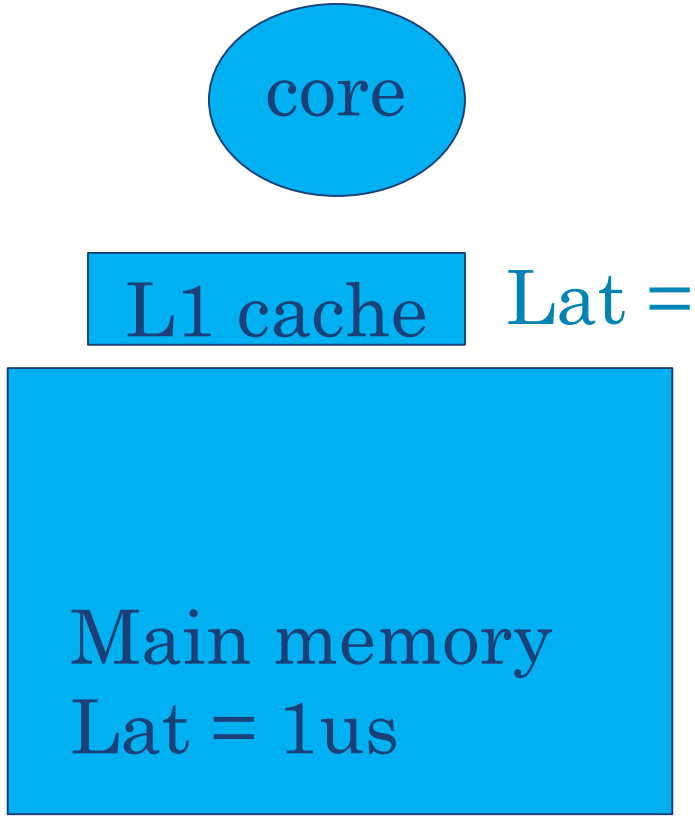


Assume 1 million requests

Total time with no cache
= 1 million * 1us = 1 sec

us = micro-second, Lat = Latency

Why memory hierarchy



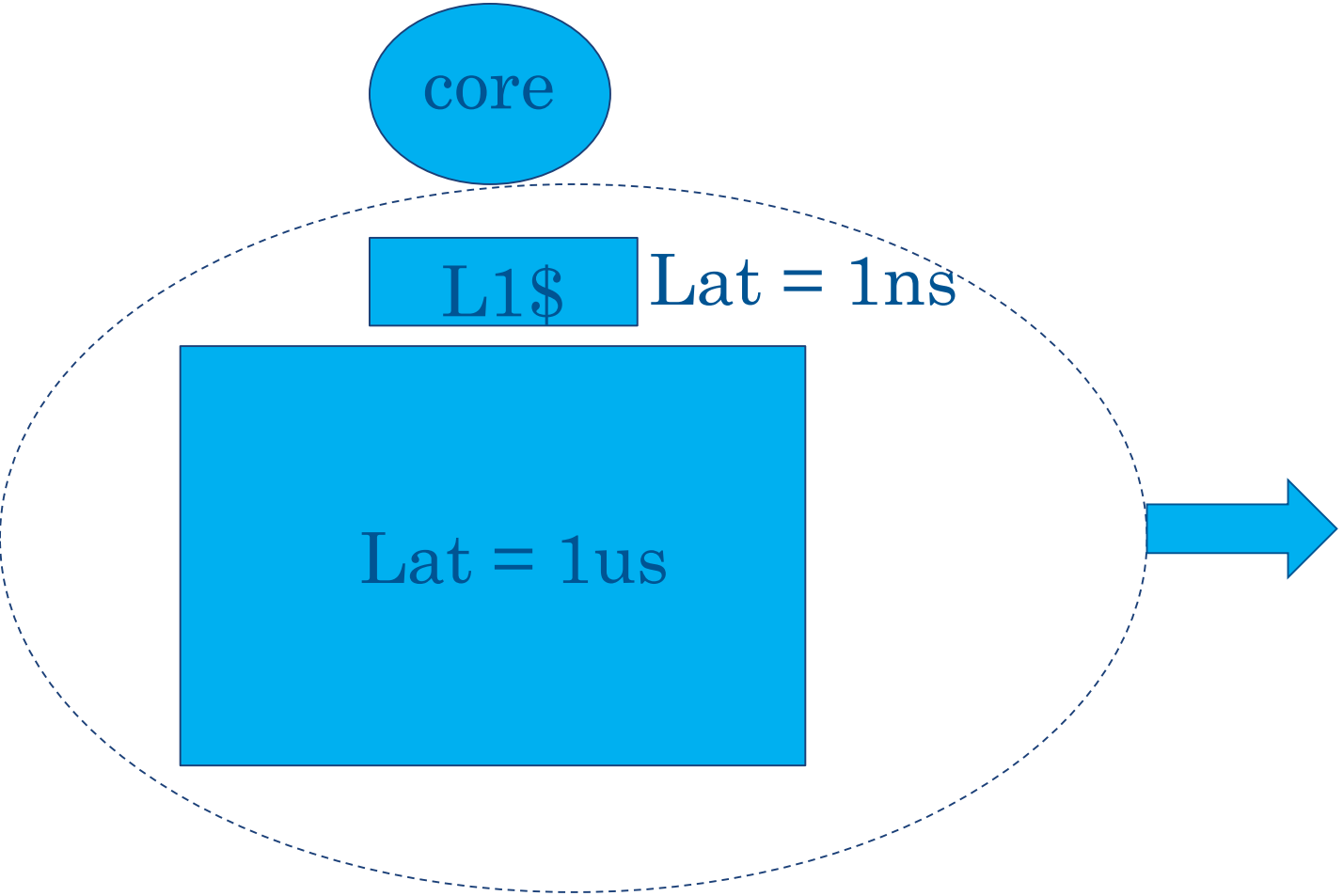
Assume 1 million requests

Total time with cache
= (HitRate*1ns + MissRate * 1us) * 10^6

With increasing hit-rate, total access time approaches that of L1 cache

Cache Hit/miss = element found or not found in cache

Why memory hierarchy



It appears to be a memory
- As fast as L1 cache
- As large as main memory

Approaches the characteristic of Ideal memory

Understanding cache block

Example 1

- A cache has 4MB cache capacity.
- Find the number of blocks in it, if block size is (a) 64B and (b) 128B
- Answer: (a) 4MB/64B = 65,536
- (b) 4MB/128B = 32,768

- ➔ On increasing the block size, we can keep less number of blocks in the cache

MB = Mega-Byte

We will now discuss an example of a small cache.
We will change the block size and show the cache operation.

Case-1: Block size is 1B

- If I access address 7, which address I get?
  Answer: 7
- If I access address 4, which address I get?
  Answer: 4

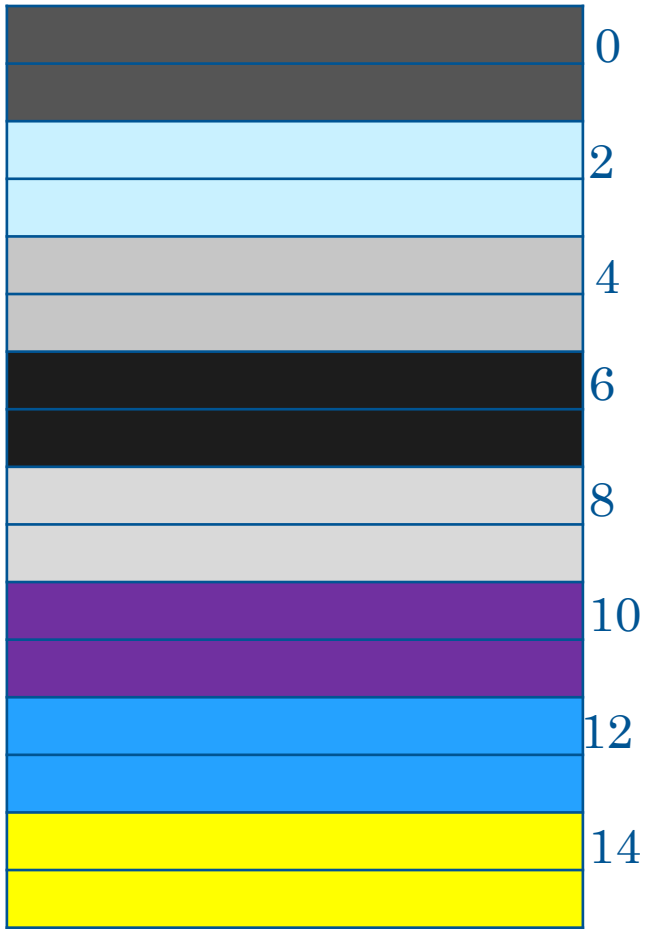
Case-1: Block size is 2B

- If I access address 7, which addresses I get?
  Answer: 6 and 7

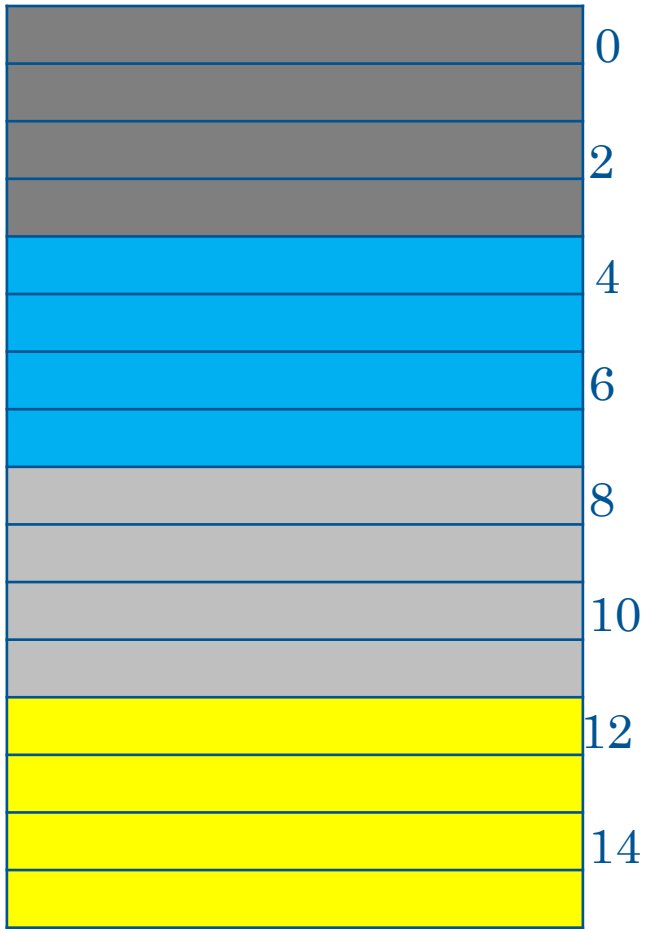
- If I access address 14, then?

  Answer: 14 and 15

Both 14 and 15 are in same block



Case-1: Block size is 4B

- If I access address 7, which addresses I get?
- Answer: 4, 5, 6 and 7

4, 5, 6 and 7 are part of the same block